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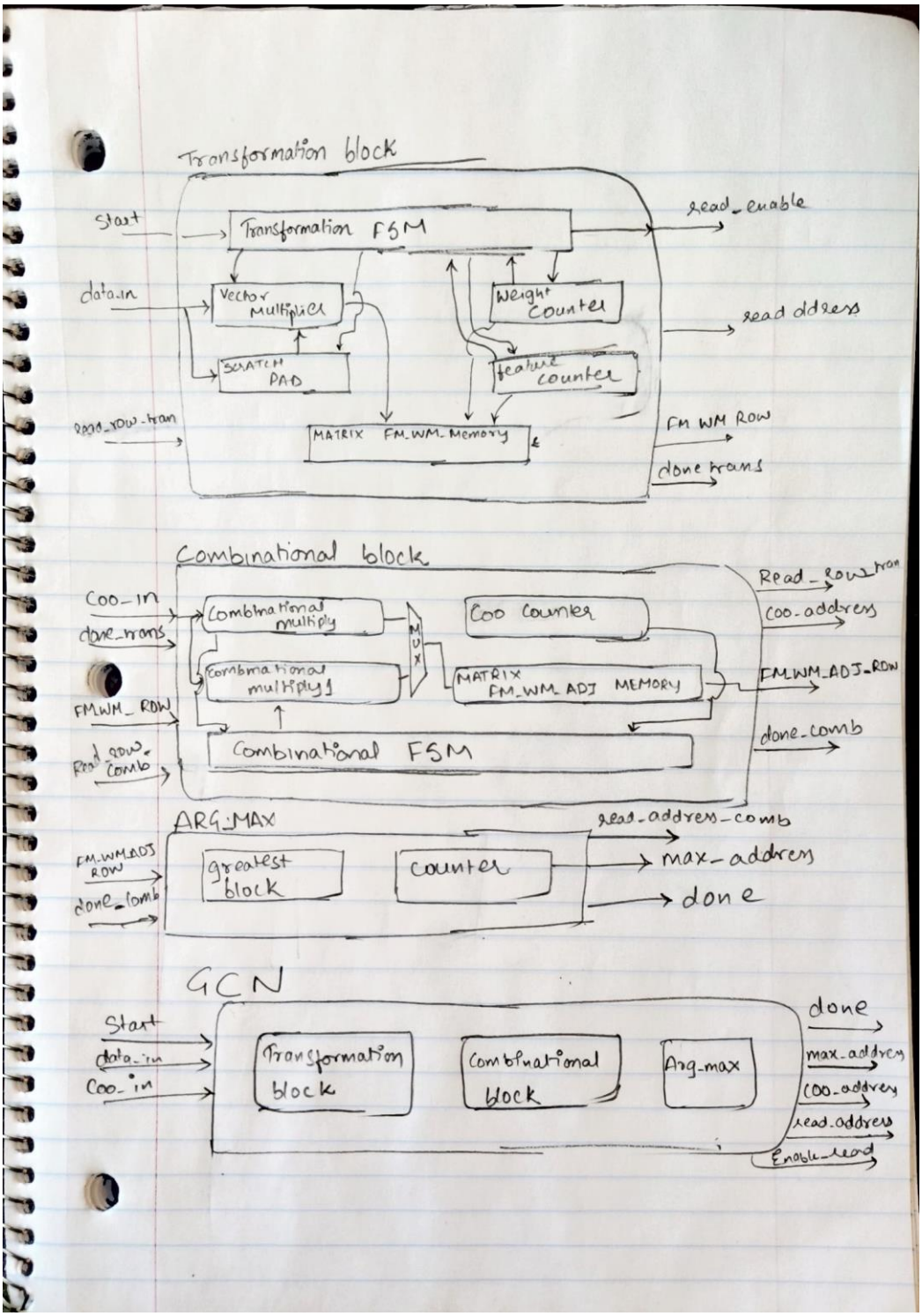
Architecture/High-level Block Diagram:

Design decisions:

- Vector_Multiplier take both the weight and the features input and while storing the weight column data in scratch repeats the feature input. In this way at almost every clock cycle it



produces the output element of the FM*WM matrix.

- Scratch_pad being the temporary memory where the input weight matrix is stored
- Matrix_FM_WM_Memory is the final output storage we use to store the FM*WM element by element.

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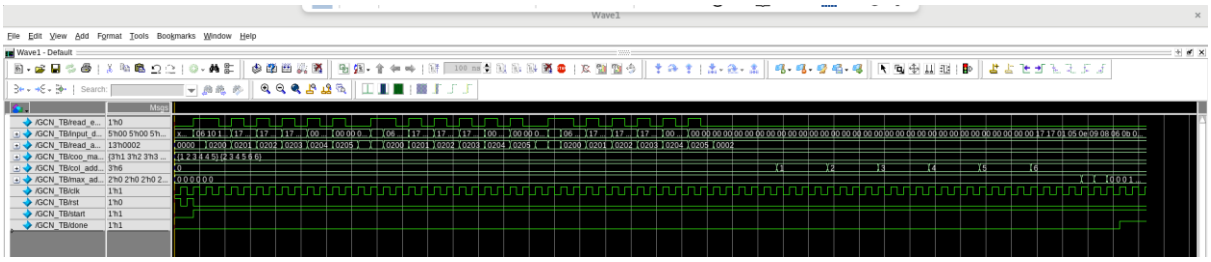
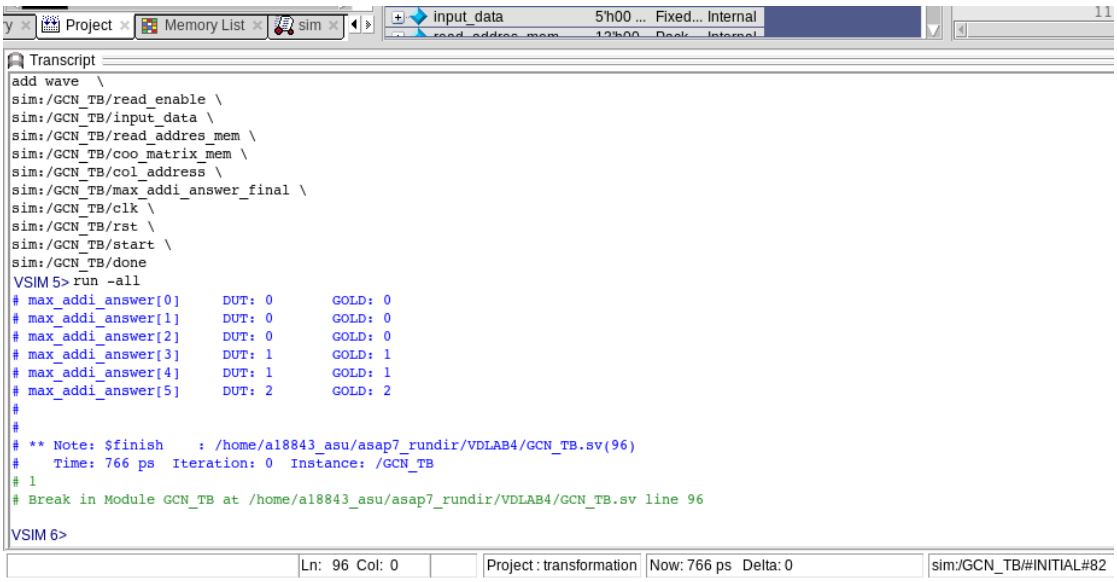
- Feature Counter: This is a rotating counter that counts up from 0 to 5 and then becomes 0.
- Weight Counter: This is a rotating counter that counts up from 0 to 5 and then becomes 0.
- The Control of when the counter should increment and when the next date is to be input and also when the combinational block should start working is decided by the Transformation FSM.
- There is also another FSM for the combinational block that is triggered to start when the done_tran signal is received from the Transformation_FSM. This Combination_FSM controls the flow of data in the Combination and also when the next argmax block should start.
- The COO_ADJ_FM_WM and the COO_ADJ_FM_WM_T take in the raw coo data and from the data of the coo it pick the row from the Matrix_FM_WM_Memory and Matrix_FM_WM_ADJ_Memory and adds them together. (This works since Coo is a sparse matrix, so we replace the output with the fm_wm_row_out).
- The Argmax block starts functioning as soon as the done_comb signal is received from the combinational_FSM and reads the data in the Matrix_FM_WM_ADJ_Memory row by row and outputs the column number of each row where the data is the largest.

Behavioral Verilog – Simulation :





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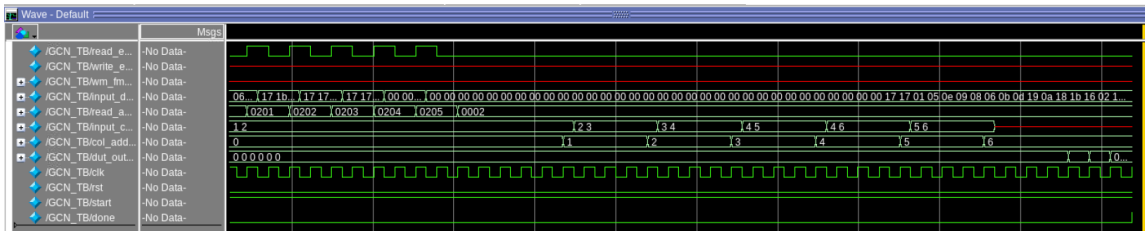
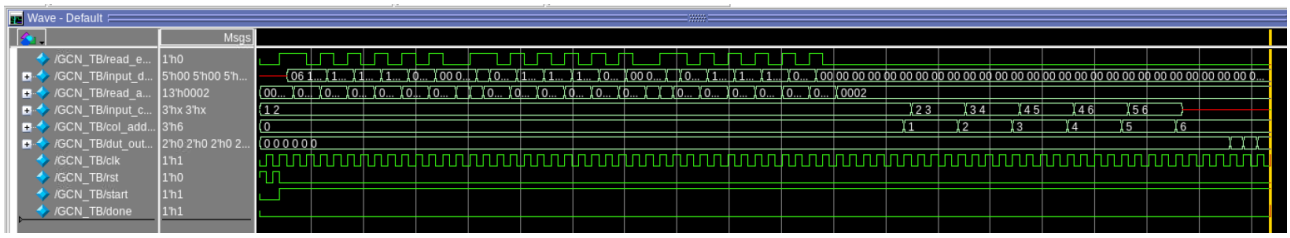
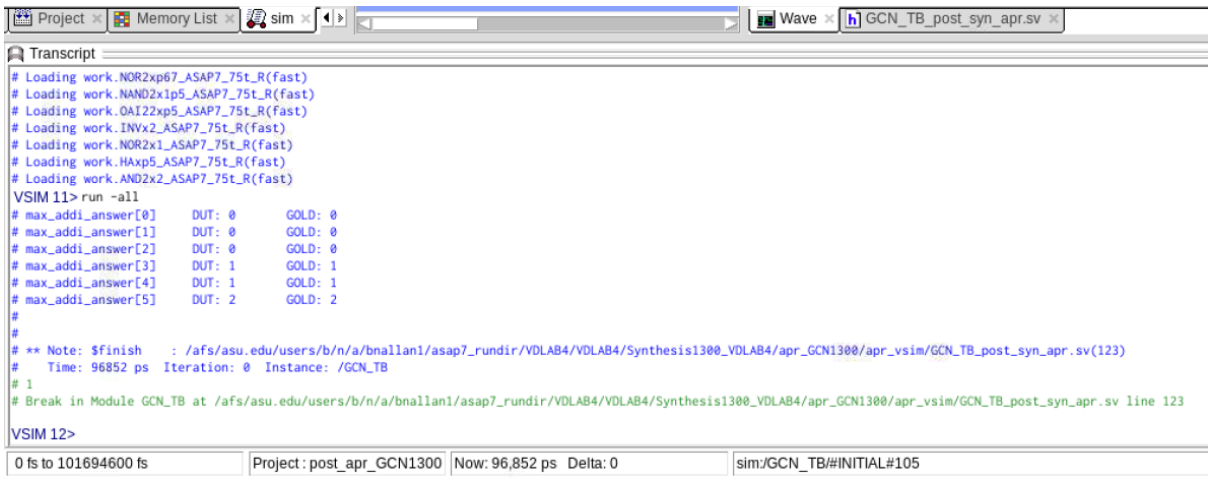
Post_Synthesis – Simulation: (Synthesis done at 1300 ps)



Milestone 2

- **Total Latency:**
  - Total Latency: 96852 ps at 0.6923 GHz (at 1300ps clock period)
  - Modelsim:





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- Power:
  - With usinf report_power cmd:

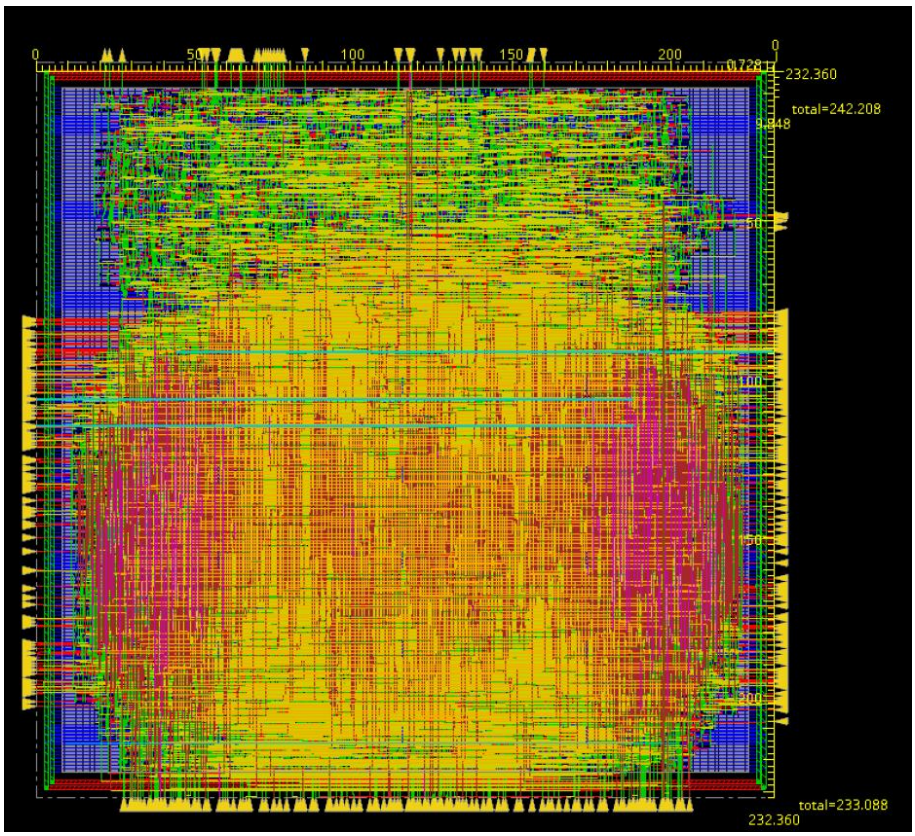
```
Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        1.46682488            39.6890%
Total Switching Power:       2.22779754            60.2792%
Total Leakage Power:         0.00117564             0.0318%
Total Power:                 3.69579806
-----------------------------------------------------------------------------------------
```

  - Using the VCD generated by Vsim to get the power:

```
Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:      183.96109460            23.5019%
Total Switching Power:     598.78802616            76.4980%
Total Leakage Power:         0.00117564             0.0002%
Total Power:               782.75029590
-----------------------------------------------------------------------------------------
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:01,
mem(process/total)=2554.87MB/2554.87MB)
```

- Area:
- Innovus density: Before filler cell insertion: 63.169%

```
+---------------------+---------+---------+---------+
Density: 63.169%
-----------------------------------------------------
Reported timing to dir timingReports
```



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```
==============================
Floorplan/Placement Information
==============================
Total area of Standard cells: 46656.000 um^2
Total area of Standard cells(Subtracting Physical Cells): 23014.705 um^2
Total area of Macros: 0.000 um^2
Total area of Blockages: 0.000 um^2
Total area of Pad cells: 0.000 um^2
Total area of Core: 46687.104 um^2
Total area of Chip: 53916.835 um^2
Effective Utilization: 1.0000e+00
Number of Cell Rows: 200
```

- Number of gates:
  - Gates = 39754
  - Cells = 17132

```
innovus 63> reportGateCount
Gate area 0.6998 um^2
[0] GCN Gates=32885 Cells=11589 Area=23014.7 um^2
innovus 64>
```

- Post_APR – DRC Check

```
innovus 60> verify_drc
 *** Starting Verify DRC (MEM: 3347.1) ***

  VERIFY DRC ...... Starting Verification
  VERIFY DRC ...... Initializing
  VERIFY DRC ...... Deleting Existing Violations
  VERIFY DRC ...... Creating Sub-Areas
  VERIFY DRC ...... Using new threading
 VERIFY DRC ...... Sub-Area: {117.504 0.000 176.256 58.752} 3 of 16  Thread : 9
 VERIFY DRC ...... Sub-Area: {0.000 117.504 58.752 176.256} 9 of 16  Thread : 2
 VERIFY DRC ...... Sub-Area: {0.000 0.000 58.752 58.752} 1 of 16  Thread : 4
 VERIFY DRC ...... Sub-Area: {117.504 117.504 176.256 176.256} 11 of 16  Thread : 5
 VERIFY DRC ...... Sub-Area: {176.256 0.000 232.272 58.752} 4 of 16  Thread : 9
 VERIFY DRC ...... Sub-Area: {58.752 0.000 117.504 58.752} 2 of 16  Thread : 4
 VERIFY DRC ...... Sub-Area: {0.000 176.256 58.752 232.128} 13 of 16  Thread : 2
 VERIFY DRC ...... Sub-Area: {176.256 117.504 232.272 176.256} 12 of 16  Thread : 5
 VERIFY DRC ...... Sub-Area: {117.504 58.752 176.256 117.504} 7 of 16  Thread : 5
 VERIFY DRC ...... Sub-Area: {0.000 58.752 58.752 117.504} 5 of 16  Thread : 4
 VERIFY DRC ...... Sub-Area: {176.256 58.752 232.272 117.504} 8 of 16  Thread : 5
 VERIFY DRC ...... Sub-Area: {58.752 58.752 117.504 117.504} 6 of 16  Thread : 4
 VERIFY DRC ...... Sub-Area: {117.504 176.256 176.256 232.128} 15 of 16  Thread : 1
 VERIFY DRC ...... Thread : 4 finished.
 VERIFY DRC ...... Thread : 7 finished.
 VERIFY DRC ...... Sub-Area: {58.752 117.504 117.504 176.256} 10 of 16  Thread : 1
 VERIFY DRC ...... Thread : 1 finished.

  Verification Complete : 0 Viols.

 *** End Verify DRC (CPU: 0:00:28.1  ELAPSED TIME: 8.00  MEM: 353.1M) ***
```

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- Post_APR – LVS Check

```
innovus 62> verifyConnectivity
VERIFY_CONNECTIVITY use new engine.

******** Start: VERIFY CONNECTIVITY ********
Start Time: Mon Dec  9 00:34:46 2024

Design Name: GCN
Database Units: 4000
Design Boundary: (0.0000, 0.0000) (232.2720, 232.1280)
Error Limit = 1000; Warning Limit = 50
Check all nets
Use 10 pthreads

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Mon Dec  9 00:34:46 2024
Time Elapsed: 0:00:01.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.9  MEM: 0.000M)
```

- GCN. geom.rpt

```
Open                                                GCN.geom.
                                                    ~/asap7_rundir/VDLAB4/VDLAB4/Synthes
1 ##############################################################
2 #  Generated by:      Cadence Innovus 17.12-s095_1
3 #  OS:                Linux x86_64(Host ID eecad5)
4 #  Generated on:      Mon Dec  9 00:33:57 2024
5 #  Design:            GCN
6 #  Command:           verify_drc
7 ##############################################################
8
9 No DRC violations were found
10
```

- GCN. conn.rpt

```
##############################################################
#  Generated by:      Cadence Innovus 17.12-s095_1
#  OS:                Linux x86_64(Host ID eecad5)
#  Generated on:      Mon Dec  9 00:34:46 2024
#  Design:            GCN
#  Command:           verifyConnectivity
##############################################################
Verify Connectivity Report is created on Mon Dec  9 00:34:46 2024



Multi-CPU acceleration using 10 CPU


Use pthread


Begin Summary
    Found no problems or warnings.
End Summary
```

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- Screenshot of your timing report for the worstcase hold and setup path

  - Setup

```
#############################################################
#  Generated by:      Cadence Innovus 17.12-s095_1
#  OS:                Linux x86_64(Host ID eecad5)
#  Generated on:      Sun Dec  8 23:47:25 2024
#  Design:            GCN
#  Command:           timeDesign -postRoute -pathReports -drvReports -slackRepor
timingReports
#############################################################
Path 1: MET Setup Check with Pin t1/TM1/VM1/multi21_reg_15_/CLK
Endpoint:   t1/TM1/VM1/multi21_reg_15_/D    (v) checked with  leading edge of
'clk'
Beginpoint: t1/TM1/SP1_memory_reg_33__0_/QN (v) triggered by  leading edge of
'clk'
Path Groups: {reg2reg}
Analysis View: default_setup_view
Other End Arrival Time          67.300
- Setup                          1.375
+ Phase Shift                  1300.000
= Required Time                1365.915
- Arrival Time                 1269.132
= Slack Time                    96.783
     Clock Rise Edge                      0.000
     + Drive Adjustment                   7.900
     + Source Insertion Delay           -77.268
     = Beginpoint Arrival Time          -69.368
     Timing Path:

+-----------------------------------------------------------------------------
```

  - Hold

```
#############################################################
#  Generated by:      Cadence Innovus 17.12-s095_1
#  OS:                Linux x86_64(Host ID eecad5)
#  Generated on:      Sun Dec  8 23:47:41 2024
#  Design:            GCN
#  Command:           timeDesign -postRoute -hold -pathReports -slackReports -nur
timingReports
#############################################################
Path 1: MET Hold Check with Pin t1/TM1/SP1_memory_reg_87__3_/CLK
Endpoint:   t1/TM1/SP1_memory_reg_87__3_/D (v) checked with  leading edge of
'clk'
Beginpoint: data_in[43]                    (v) triggered by  leading edge of '@'
Path Groups: {clk}
Analysis View: default_hold_view
Other End Arrival Time          12.044
+ Hold                          16.636
+ Phase Shift                    0.000
= Required Time                 28.690
  Arrival Time                  28.700
  Slack Time                     0.010
     Clock Rise Edge                      0.000
     + Input Delay                        0.000
     = Beginpoint Arrival Time            0.000
     Timing Path:
```